

A Literal Counterexample to the Unit-Ball Extension Clause in Problem 22

Packet for arXiv:2602.15525

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Source Question

The source is S.A. Bogaty and A.A. Tuzhilin, *Mazur–Ulam Theorem With Gromov–Hausdorff Distance*, arXiv:2602.15525. Problem 22 asks:

The condition (5') from Theorem 16 and Problem 19 can be connected as follows.

Problem 22. *Is it true that if for Banach spaces V and W there is an isometric embedding $f : B_V \rightarrow W$, then there is a (linear) isometric embedding $F : V \rightarrow W$ (such that $F|_{B_V} = f$)?*

For a Banach space V , let $Q(V)$ denote the space of all Banach spaces isomorphic to

In transcription, the problem asks whether, if Banach spaces V and W admit an isometric embedding $f : B_V \rightarrow W$, then there is a linear isometric embedding $F : V \rightarrow W$ such that $F|_{B_V} = f$.

Answer

The literal extension statement is false. There are Banach spaces V, W and an isometric embedding $f : B_V \rightarrow W$ which cannot be the restriction of any linear map $V \rightarrow W$, hence cannot be the restriction of a linear isometric embedding.

Proof Intuition

The obstruction is that an isometric embedding of a ball need not remember the linear oddness of the ambient space. The map $v \mapsto (v, \|v\|)$ places the unit ball on the graph of the norm in an ℓ_∞ -sum. Distances are unchanged because the norm coordinate is 1-Lipschitz. But the graph is not centrally symmetric: u and $-u$ both have second coordinate 1. A linear extension would have to send $-u$ to the negative of the image of u , which is impossible.

Theorem 1. *Let V be any nonzero real Banach space. Put*

$$W = V \oplus_\infty \mathbb{R}, \quad \|(x, t)\|_W = \max\{\|x\|_V, |t|\}.$$

Then

$$f : B_V \rightarrow W, \quad f(v) = (v, \|v\|_V),$$

is an isometric embedding, but no linear map $F : V \rightarrow W$ satisfies $F|_{B_V} = f$.

Proof. For $v, w \in B_V$,

$$\|f(v) - f(w)\|_W = \max\{\|v - w\|_V, \left| \|v\|_V - \|w\|_V \right|\}.$$

By the reverse triangle inequality,

$$\left| \|v\|_V - \|w\|_V \right| \leq \|v - w\|_V.$$

Thus $\|f(v) - f(w)\|_W = \|v - w\|_V$, so f is an isometric embedding of B_V into W .

Choose $u \in V$ with $\|u\|_V = 1$. If a linear map $F : V \rightarrow W$ satisfied $F|_{B_V} = f$, then

$$F(u) = f(u) = (u, 1) \quad \text{and} \quad F(-u) = f(-u) = (-u, 1).$$

Linearity also gives

$$F(-u) = -F(u) = (-u, -1),$$

which contradicts $F(-u) = (-u, 1)$. Therefore no such linear extension exists. $\square$

Concrete One-Dimensional Instance

Taking $V = \mathbb{R}$ and $W = \ell_\infty^2$, the map

$$[-1, 1] \ni t \mapsto (t, |t|)$$

is an isometric embedding, since

$$\max\{|s - t|, \left| |s| - |t| \right|\} = |s - t|$$

for all $s, t \in [-1, 1]$. It cannot be the restriction of a linear map $\mathbb{R} \rightarrow \ell_\infty^2$, because it sends both 1 and -1 to points with second coordinate 1.

Scope and Limitations

This packet refutes the literal parenthetical extension requirement $F|_{B_V} = f$. It does not refute the weaker reading that asks only whether W contains some linear isometric copy of V whenever B_V embeds isometrically into W . In the construction above, the weaker conclusion is true, because $v \mapsto (v, 0)$ is a linear isometric embedding of V into $V \oplus_\infty \mathbb{R}$.

The same source paper recalls closely related nonlinear norm-graph constructions before stating Problem 22. Therefore the human review question is partly editorial: this should be counted either as a counterexample to Problem 22 as written, or as an internal wording issue rather than as a new answer to the weaker intended problem.

Verification and Novelty Check

The proof uses only the reverse triangle inequality and linearity. The included scalar sanity script checks the one-dimensional instance on a finite grid; it is not used as proof.

Local run indexes were searched for arXiv:2602.15525, `pr:ball`, `unit ball embedding`, and exact phrases involving $F|_{B_V} = f$. Web searches on 2026-06-21 for exact variants of Problem 22 found the source arXiv record and no separate later answer. The novelty confidence is intentionally limited because the counterexample is a one-line consequence of a construction already adjacent in the source paper.

References

- [1] S.A. Bogaty and A.A. Tuzhilin, *Mazur–Ulam Theorem With Gromov–Hausdorff Distance*, arXiv:2602.15525, 2026.